

1. Let $S = \left\{ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10 \right\}$ and $\beta = \sum_{x \in S} \tan^2 \left(\frac{x}{3} \right)$, then $\frac{1}{6} (\beta - 14)^2$ is equal to

[2023 (10 Apr Shift 2)]

- (1) 16
(2) 8
(3) 64
(4) 32

ANSWER KEYS

1. (4)

1. (4)

Given,

$$9^{1-\tan^2 x} + 9^{\tan^2 x} = 10$$

Now let $9^{\tan^2 x} = t$, then above equation will be,

$$\frac{9}{t} + t = 10$$

$$\Rightarrow t^2 - 10t + 9 = 0$$

$$\Rightarrow t = 9 \text{ or } t = 1$$

So, when $9^{\tan^2 x} = 9 \Rightarrow \tan^2 x = 1$

$$\Rightarrow \tan x = \pm 1 \Rightarrow x = \pm \frac{\pi}{4}, \text{ as given } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Now when $9^{\tan^2 x} = 1$

$$\Rightarrow \tan^2 x = 0 \Rightarrow x = 0$$

$$\text{Hence, } \beta = \sum_{x \in S} \left(\frac{x}{3}\right) = \tan^2\left(\frac{0}{3}\right) + \tan^2\left(\frac{\pi}{12}\right) + \tan^2\left(\frac{-\pi}{12}\right)$$

$$= 0 + 2\left(2 - \sqrt{3}\right)^2$$

$$= 14 - 8\sqrt{3}$$

$$\text{So, the value of } \frac{1}{6}(\beta - 14)^2 = \frac{64 \times 3}{6} = 32$$

Hence this is the correct option.